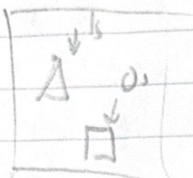
HW - 142 Econ

1) a) C

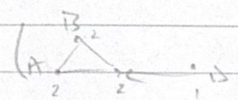
b) D



c) D

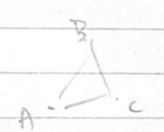
d) A

2) a) FALSE (A ——— B Diam = 1, A ——— B ——— C Diam = 2)

b) FALSE ( → 7/4 avg,  → 6/3 = 2 avg)

7/4 < 2

c) TRUE (once you have a minimum t, there will always be a way to increase to t+1, which repeats. For example



, t = 1, but you can also go from each point to each other with t = 2 +)

d) TRUE (As long as everything is connected then there is a max number of steps to take to get from one point to another)

3)

	Neighbors	Possible connections	Actual	CC
A	4	6	1	1/6
B	2	1	1	1
C	2	1	0	0
D	2	1	0	0
E	1	0	0	N/A
F	2	1	0	0

a) Diameter = 3

b) →

Max: B = 1

Min: C, D, F = 0

c) AE is a bridge because breaking the connection splits the network into 2 distinct parts ;

AC, CD, and FD are local bridges because they don't share a common neighbor.

d) FB edge has highest network overlap because A is a common neighbor.

4) a) E[Links] = # of possible links · prob. of link = $\boxed{n_a \cdot n_b \cdot \frac{p}{2}}$

b) P(A-B) < P(A-A), P(AB) < P(B-B) ∴ There will be homophily. It occurs when similar nodes are more likely to be linked than different. and $p/2 < p$.

Now $n_a = n_b = n$. $(n-1)p = 8$

c) Expected degree : $(n-1)p = 8$ so it will follow a poisson distribution with $\lambda = 8$

$\lambda = n-1(p(n))$
$= n-1\left(\frac{8}{n-1}\right)$

$\boxed{8^d e^{-8}/d!}$

a linked to
↓ b

d) When n is large, Expected degree $= (n-1)p + \frac{np}{2}$
$= 8 + \frac{n}{n-1} \cdot 4$

$\lim_{n \to \infty} \frac{n}{n-1} \cdot 4 = 4$

∴ $\lambda = 8+4 = 12$ ⟶ $\boxed{12^d e^{-12}/12!}$